SaayaShift - Heat-Aware Shift Planner

HackInsight 2026 | Karachi-first | Open-source | MIT

Problem & User

User: Supervisor scheduling outdoor field crews in Karachi. Generic weather dashboards do not turn forecasts into explainable shift options. Dawn (Jun 2023, Jun 2024) and Climate Rights International (2025) document occupational heat risk for outdoor workers in Karachi. These are qualified problem evidence; individual outcomes are not predicted.

Data Sources & Warehouse

Source 1: Open-Meteo Historical Forecast + Forecast API (CC BY 4.0). 26,472 hourly rows covering 2023-01-01 to 2025-12-31 plus 7-day live forecast. Grid cell 24.8607N 67.0011E. Provenance in manifest.csv (SHA-256, fetch time, licence).

Source 2: OpenStreetMap via Overpass API (ODbL). Karachi area snapshot. Geographic context only (not safety measurements). STATUS: BLOCKED - all Overpass mirrors unreachable from event network. Stub clearly labeled. Layers: Bronze (raw JSON) / Silver (typed, deduped, imputed) / Gold (validated, analysis-ready).

Data Quality Agent

Gate 1 (Silver): schema check, type cast, dedup with forecast winning on overlap, range check, linear interpolation for gaps up to 3h. Gate 2 (Gold): completeness <2%, consistency (apparent temp range), freshness, drift detection (3-sigma). Fail = alert + keep last-good Gold. Every decision logged to data/silver/dq_log.csv with rule, action, row count, reason.

Model & Evaluation

Target: hourly apparent temperature (C). Frozen baseline: lag-24h (H minus 24h), declared before results viewed. Train: 2023-2024 | Test: 2025 holdout (chronological, untouched).

Baseline MAE: 1.0302 C | LightGBM MAE: 0.2246 C | Improvement: 0.8056 C | Beats baseline: YES | 90% PI coverage: 86.04%. Slices by month and hour-of-day in docs/model_card.md. Limit: single grid cell; model-grid estimates, not street-level observations.

Insight Agent (RAG)

Corpus: OSHA, ILO, WHO heat-safety guidance (openly licensed). Hybrid keyword + vector search (sentence-transformers all-MiniLM-L6-v2 + FAISS). Guardrails: numbers from Gold SQL only (not LLM memory); abstains when relevance < 0.40; no PII in prompts; cites title / publisher / URL / section / chunk_id per answer. Does not diagnose illness, certify shifts, or replace official guidance.

Pipeline & Dashboard

8-step hourly flow: Fetch -> Bronze -> Gate 1 -> Gate 2 -> Retrain (drift) -> Re-index RAG -> Briefs -> Dashboard refresh. Orchestration: pipeline/scheduler.py (no admin required). Evidence window: Tue 20 Oct 16:00 PKT - Wed 21 Oct 16:00 PKT.

Gradio dashboard (5 tabs): Shift Planner / Forecast View / Map Context / Safety Q&A / Pipeline Status. Shows data freshness, DQ gate status, last successful run, model MAE vs baseline. Attribution: Open-Meteo (CC BY 4.0) and OSM contributors (ODbL) visible.

Pakistan Impact & Responsible AI

Direct beneficiaries: outdoor supervisors and field crews in Karachi facing occupational heat risk. No personal data collected. No individual illness predicted. All thresholds from cited WHO/ILO/OSHA guidance. Data limitations labeled. AI use logged in AI_Use_Log.md. City-configurable for future expansion.